

By defn. of negation, $\neg(\neg(\exists x(C(x) \wedge F(x))))$ is False. By double-negation law, $\exists x(C(x) \wedge F(x))$ is False.

By defn. of AND, $\exists x \exists y \exists z \exists w ((C(x) \wedge F(x)) \wedge (x \neq y) \wedge (C(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (C(z) \wedge T(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (C(w) \wedge Se(w)))$ is False.

By defn. of OR, $\exists x \exists y \exists z \exists w ((M(x) \wedge F(x)) \wedge (x \neq y) \wedge (M(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (M(z) \wedge T(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (M(w) \wedge Se(w))) \vee ((C(x) \wedge F(x)) \wedge (x \neq y) \wedge (C(y) \wedge S(y)) \wedge (x \neq z) \wedge (y \neq z) \wedge (C(z) \wedge T(z)) \wedge (x \neq w) \wedge (y \neq w) \wedge (z \neq w) \wedge (C(w) \wedge Se(w)))$ is False.

17) Express each of these system specifications using predicates, quantifiers and logical connectives, if necessary.

a) Every user has access to exactly one mailbox:

$A(x, y)$: User x has access to mailbox y

Domain of x : Set of users

Domain of y : Set of all mailboxes

$\forall x \exists y (A(x, y) \wedge \forall z ((y \neq z) \Rightarrow \neg A(x, z)))$

b) There is a process that continues to run during all error conditions only if the kernel is working correctly

$R(x, y)$: Process x runs during error condition y

Domain of x : Set of all processes

Domain of y : Set of all error conditions.

$\exists x \forall y (R(x, y)) \Rightarrow \exists z K(z)$

$K(z)$: Kernel z is working correctly. The domain is the set of all kernels.

c) All users on the campus network can access all websites whose url has a .edu extension

$A(x, y, z)$: User x accesses website y with a url z .